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Tech Bytes | Europe

Space Technology – Orbital Computing

Space technology offers a transformative solution by building compute capacity into space, leveraging abundant solar energy and radiative cooling. Advances in launch economics, thermal management, and power systems offer unique orbital infrastructure investment opportunities.

What is orbital compute? These are racks in space (not floating giant data centers) with solar arrays kept in a sun synchronous orbit and a cooling radiator behind it constantly in the shade. It is a virtual data centre where racks are linked using lasers travelling through vacuum, which are technologies widely used in modern satellites. These platforms can drive higher cost efficiencies, alleviate terrestrial constraints, reduce environmental impact, and enable new capabilities for latency-sensitive and security-critical applications.

Why put data centers in space? Nearly half of today's cost of building a 1GW AI data center is spent on infrastructure, rather than compute**.** As reusable launch continues to drive costs lower, orbital platforms could replace a significant portion of this terrestrial infrastructure with solar-powered systems in space, potentially bringing orbital compute economics closer to terrestrial AI infrastructure over time. The space economy is poised to reach US$1.8tr by 2035e, according to *World Economic Forum,* led by the next generation of satellite networks developed by companies like SpaceX's Starlink, Amazon's Kuiper, Eutelsat's OneWeb, and China's Guowang exploring a space-based scalable AI infrastructure.

The space technology supply chain is a highly specialized ecosystem combining new compute architectures suited to the space environment with a mechanical design in which solar power collection, compute, and thermal management are tightly integrated. Its commercial mega-constellations and reusable rocket technology operates at the intersection of semiconductors, communications, aerospace, defense, and advanced AI infrastructure.

Investment implications. We curated a list of tech companies at the forefront of space technology, leading in innovation and market presence. Building orbital compute takes a tremendous amount of radiation-tested components with most of the value in (1) optical/RF, (2) power/cooling and (3) semiconductor chips. In Europe, we favour **STMicro** and **Infineon** as companies providing these enabling technologies and which stand to benefit; understanding which capabilities are advancing fastest, ecosystem readiness and adoption timelines will dictate commercial traction.

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Executive summary

Orbital compute is an emerging AI infrastructure theme at the intersection of semiconductors, satellite systems, launch, thermal management, optical communications and autonomous operations. We do not expect it to replace terrestrial hyperscale data centers this cycle. The more realistic near-term opportunity is orbital edge AI: satellites process imagery, sensor data and inference workloads in orbit before sending only useful outputs back to Earth. The longer-term bull case is a distributed AI infrastructure layer in space.

Why now? Four structural trends are making orbital compute increasingly credible. AI data centers are running into power, land, water and permitting constraints; reusable launch is reducing the cost of putting mass into orbit; optical satellite networking is evolving toward distributed compute architectures; and the volume of space-generated data continues to grow. Together, these trends strengthen the case for processing some workloads in orbit. However, orbital compute does not eliminate infrastructure constraints; it replaces terrestrial bottlenecks with new engineering challenges around launch cost, thermal management, radiation tolerance, optical bandwidth, orbital debris and autonomous operations.

Exhibit 1: Potential Benefits and Key Challenges

Potential Benefits	Rationale	Key Challenges	Rationale
Power	Near-continuous solar exposure in dawn–dusk SSO; no terrestrial grid constraints.	Radiation	Requires radiation-tolerant chips, ECC and shielding.
Thermal	No air or water cooling, but heat must still be rejected via heat pipes and radiators.	Repair / Refresh	On-orbit maintenance is difficult; AI hardware refresh cycles are short.
Latency	LEO enables low-latency connectivity (~10–30 ms RTT).	Optical networking	AI-scale distributed compute needs far higher inter-satellite bandwidth than today's satellite networks.
Scalability	Reusable launch vehicles could lower deployment cost over time.	Orbital debris / security	Collision avoidance, cyber risk and physical protection become core operating requirements.

Source: Morgan Stanley Research

Where we are today? Orbital compute remains in the early stages, but it is no longer only a research concept. SpaceX Starmind is the clearest public blueprint for a scalable orbital AI infrastructure system, while Starcloud and NVIDIA are exploring compute hardware, orbital platforms and space-native AI workloads. China is also moving from concept to deployment: Adaspace reportedly launched the first 12 satellites of its "Three-Body Computing Constellation" in May 2025, with a stated long-term plan for 2,800 satellites and onboard AI processing. The near-term commercial use case remains orbital edge AI – processing satellite imagery, sensor data and inference workloads in orbit – while large-scale, space-based AI infrastructure remains a longer-term option.

Economics: We use SpaceX initiation estimates as the economics benchmark and frame orbital compute around cost per watt. In orbit, terrestrial power, land and water constraints are replaced by launch cost, satellite hardware, solar arrays, radiators, radiation tolerance and autonomous operations. Our Embodied AI/Robotics team, led by Adam Jonas, provides the most detailed benchmark: orbital compute capex/W is modeled to decline from roughly US$60/W in 2030 to US$32/W in 2031, US$15/W in 2035 and US $9/W by 2040, driven by lower launch cost, cheaper satellite hardware and improved compute payload economics. On a five-year useful-life basis, the 2031 estimate implies roughly US$6.5/W/year, close to the current industry Blackwell benchmark of US$6.8/W/year. For details, please see the SpaceX initiation – "AI's Final Frontier".

Stock implications

Orbital compute extends the existing AI hardware stack where additional technologies are required to enable reliable performance in space.

- The key investment exposure sits in RF / phased-array hardware (**STM, Qorvo** and **Amphenol**), optical connectivity (**Coherent/Lumentum)**, and power and thermal systems (**GS Yuasa**, **Lite-On** and **LG Energy Solution**), which form the enabling layer around the core compute payload.
- In parallel, **Infineon**, **Analog Devices**, **Microchip** and **Shanghai Fudan** provide radiation-tolerant, high-reliability semiconductors essential for stable operation in orbit, with Infineon positioned in space-grade power management.
- Core AI silicon remains necessary for compute payloads (**NVIDIA**, **AMD** – AMD Versal AI Core adaptive system-on-chips (SoCs) in satellites to handle critical AI processing and wireless data acceleration; **Broadcom**, **Micron**, **TSMC**, **SK hynix** and **Samsung** – developed specialized Exynos modem chips (Exynos 5400) that use machine learning to optimize communication with low Earth orbit satellites, particularly for their Direct to Cell networks).

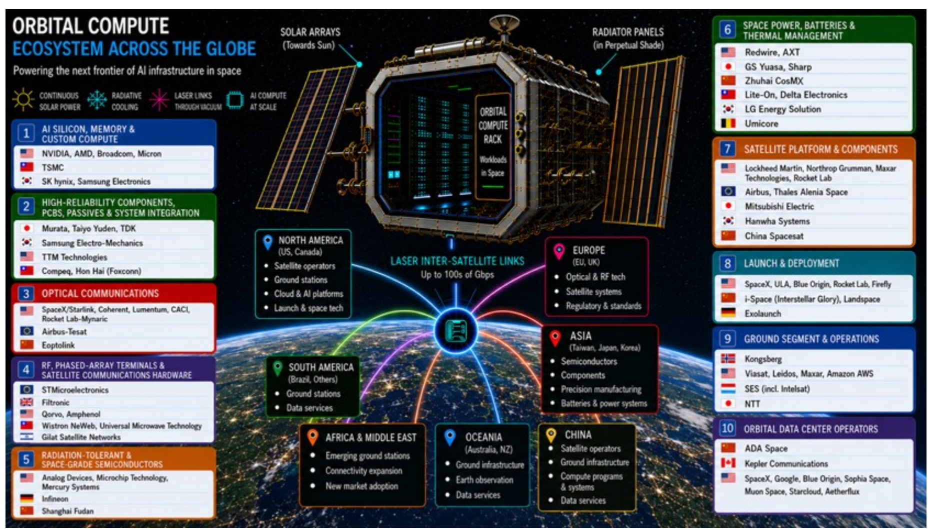
Within our European Semiconductors coverage, both **STMicroelectronics** and **Infineon** stand out as key enablers. Both are key suppliers of radiation-hardened semis that are essential for power management ICs, flight computers, memory for critical systems, FPGAs controlling the spacecraft and communications and attitude-control electronics – for instance, rad hard power MOSFETs, solid state relays and space shottky diodes. STMicro recently held a specific low Earth orbit event providing more granular detail on technology positioning and the bill of materials (see our note, *LEO opportunity enters orbit*). The company is guiding to "well above US$3bn cumulative sales over the next 3 years" and separates the LEO opportunity into 3 brackets: (i) Broadband, (ii) direct-to-cell and (iii) orbital data centre. User terminals were highlighted by management as the largest revenue contributor, with the company calling out its BiCMOS technology and PLP (panel level packaging) as enabling competitiveness in user terminal front-end modules. We model user terminals, satellite and gateway sales separately, estimating US$546mn user terminal sales in FY26, 67% of LEO sales. All in, we see STM's LEO sales ramping from US$815mn FY26 to US$1.8bn FY28. Our sense is that this could be conservative, as satellite launches may ramp at a larger size than we are modelling, especially as more new players enter the market.

Exhibit 2: Orbital compute technology supply chain

Company	Ticker	Region	Rating	Last Close (LC)	Market Cap ($mn)	Key orbital read-through
1. AI silicon, memory & custom compute						
NVIDIA	NVDA	US	OW	195	4,714,886	Space-1 / Vera Rubin, Jetson Orin and IGX Thor make NVIDIA the default orbital AI accelerator reference.
AMD	AMD	US	EW	518	844,357	Space-grade Versal adaptive SoCs and data acceleration for onboard processing payloads.
Broadcom	AVGO	US	OW	360	1,714,870	Custom silicon and networking read-through for AI payloads, ground gateways and high-bandwidth systems.
Micron	MU	US	OW	976	1,101,788	HBM / DRAM supply exposure for AI compute payloads and resilient orbital memory.
TSMC	2330 TT	Taiwan	OW	2,445	1,982,637	Leading-edge foundry for orbital AI silicon; any scaled compute layer ultimately depends on advanced logic supply.
SK hynix	000660 KS	Korea	OW	2,425,000	1,123,802	HBM is a binding constraint for any gigawatt-scale deployed compute architecture.
Samsung Electronics	005930 KS	Korea	OW	309,500	1,317,860	HBM / DRAM exposure; additional satellite communications optionality through modem / RF ecosystem.
2. High-reliability components, PCBs, passives & system integration						
Murata	6981 JP	Japan	OW	11,080	125,051	Space-grade MLCC leader; high-reliability passives for satellites, power modules and payload electronics.
Taiyo Yuden	6976 JP	Japan	UW	20,560	15,953	High-reliability MLCCs and passive components for harsh-environment electronics.
TDK	6762 JP	Japan	OW	3,699	43,533	Passives and electronic components for space, power and satellite systems.
Meiko Electronics	6787 JP	Japan	EW	29,330	4,668	Strong market position in PCB antennas for LEO satellite communication receivable antennas, and could potentially expand into satellite-grade PCB
Samsung Electro-Mechanics	009150 KS	Korea	OW	1,989,000	98,314	MLCC / substrates exposure for satellite electronics and power modules.
TTM Technologies	TTMI	US	NC	156	16,199	PCBs for high-reliability aerospace, defense and compute payload electronics.
Hon Hai	2317 TT	Taiwan	OW	241	105,488	EMS / system-integration optionality plus direct LEO satellite communications hardware validation through Foxconn PEARL satellites.
Compeq	2313 TT	Taiwan	NC	230	8,794	Advanced PCB exposure to SpaceX satellite, ground receiving stations and terminals.
Unitech	2367 TT	Taiwan	NC	55	1,215	Advanced PCB exposure to SpaceX user terminals/ground stations.
3. Optical links, photonics & terminals						
Coherent	COHR	US	EW	333	65,218	Laser optics and photonics read-through for optical inter-satellite links and ground optical systems.
Lumentum	LITE	US	EW	728	56,663	Laser optics and photonics exposure to high-bandwidth optical networking.
CACI	CACI	US	NC	503	11,108	Optical communications terminals and space communications systems; closer to the orbital laser-link bottleneck than generic IT services.
Eoptolink	300502 CH	China	OW	526	108,158	Optical transceiver read-through for high-bandwidth orbital networking.
4. RF, phased-array terminals & satellite communications hardware						
STMicroelectronics	STM	Europe	NC	68	60,749	RF front-end / BiCMOS and space-grade electronics exposure for Starlink-style phased-array antennas, LEO terminals and satellite communications.
Filtronic	FTC LN	Europe	NC	2	699	High-frequency Solid State Power Amplifiers (SSPAs) installed in ground stati...
Gilat Satellite Networks	GILT	Israel / US	NC	13	974	Ground systems and satellite communications terminal exposure.
Qorvo	QRVO	US	EW	88	7,725	RF products and semis for satellite communications and phased-array systems.
Amphenol	APH	US	NC	165	202,484	High-reliability connectors, interconnects and cable assemblies for satellites, ground infrastructure and launch systems.
MediaTek	2454 TT	Taiwan	OW	4,195	209,370	5G NTN / satellite IoT chipset exposure; relevant to direct-to-device and satellite communications hardware rather than AI compute payloads.
WinSemi	3105 TT	Taiwan	UW	411	5,448	Taiwan compound-semiconductor foundry with GaAs / GaN RF exposure; potential beneficiary of LEO satellite communications, higher-frequency
Wistron NeWeb	6285 TT	Taiwan	NC	258	3,902	LEO ground terminals, phased-array antennas and satellite communications equipment.
Universal Microwave Technology	3491 TT	Taiwan	NC	1,340	2,883	RF / microwave modules for satellite terminals and phased-array communications.
Rapidtek Technologies	6980 TT	Taiwan	NC	136	123	RF, satellite applications and ground-to-space / LEO terminal exposure; potential Taiwan orbital-communications overlay name.
Intellian Technologies	189300 KS	Korea	NC	73,700	493	Ground-station antennas and satellite terminal exposure.
5. Radiation-tolerant / space-grade semiconductors						
Infineon	IFX-DE	Europe	NC	77	115,294	Radiation-hardened power management, memory/RF and high-reliability power conversion for satellite platforms and compute payload power rails.
Analog Devices	ADI	US	OW	377	183,710	Radiation-hardened signal-chain, power and control electronics for space-qualified systems.
Microchip Technology	MCHP	US	EW	85	45,882	Rad-hardened semiconductors, FPGAs and space-qualified control electronics.
Mercury Systems	MRCY	US	NC	126	7,578	RF products, semiconductors and mission electronics for space and defense platforms.
Shanghai Fudan	1385 HK	China	OW	31	3,247	China space-grade FPGA supplier, benefiting from China's low earth orbit satellite launches
6. Space power, batteries & thermal management						
Redwire	RDW	US	NC	11	2,701	Deployable solar arrays (ROSA), structures and space infrastructure hardware; higher-beta pure-play exposure.
AXT	AXTI	US	NC	57	3,704	Germanium substrate exposure with China supply-chain risk framing.
GS Yuasa	6674 JP	Japan	OW	6,737	4,190	Space lithium-ion cells and commercial satellite battery exposure.
Sharp	6753 JP	Japan	NC	639	2,573	Space-grade triple-junction solar cell exposure.
Delta Electronics	2308 TT	Taiwan	OW	2,075	168,540	Power conversion, DC/DC modules and power-management exposure for orbital compute power rails and ground infrastructure.
Lite-On	2301 TT	Taiwan	EW	219	15,528	Power supply and power-conversion exposure for LEO satellite / ground infrastructure power systems.
LG Energy Solution	373220 KS	Korea	EW	362,500	55,499	Battery exposure for satellite and space-power applications.

Source: Factset, Company Data, Morgan Stanley Research

Exhibit 3: Orbital Compute Ecosystem



Note: Meant to give a representative snapshot. Does not capture 100% of exposed players. Source: Morgan Stanley Research

What is orbital compute?

A rack of compute in space. Orbital compute consists of placing meaningful compute payloads in space – AI accelerators, CPUs, memory, storage and networking – on satellites or orbital platforms with the aim to process data in orbit and eventually create a distributed compute layer around Earth. The commercial starting point is not a 1GW AI training campus in space but a more realistic path in orbital edge computing such as satellites process images, sensor data and inference workloads before sending only useful outputs back to Earth. This reduces latency and downlink bandwidth and aligns with demand for Earth observation, defense, maritime, disaster monitoring and autonomous spacecraft.

Putting the cloud in the cloud. The easy part is the chips, but the moment we enter space, it is brutal to silicon, constantly bombarding it with particles and radiation that can flip bit in memory, corrupt calculations and slowly degrade performance. This requires shielding, error-correction mechanisms and system redundancy. Encouragingly, commercial AI GPUs have already been demonstrated for inference in orbit, with Starcloud 1 deploying NVIDIA H100 GPUs in space.

How will it evolve?

Where are we today? Orbital compute is moving past the early proof-of-concept stage but not yet at commercial scale deployment. Today's activity is focused on proving that high-performance compute, optical links and thermal systems can work reliably in orbit. The near-term opportunity is orbital edge AI and the long-term vision is a distributed AI infrastructure layer in space.

Stage 1: Orbital edge computing

The near-term opportunity is processing data where it is generated. Satellites become edge AI nodes that analyze images, sensor feeds and inference workloads onboard, then downlink only useful outputs. This reduces latency and bandwidth demand, and is most relevant for Earth observation, defense, maritime monitoring, disaster response and autonomous spacecraft. The engineering requirement is not a full AI data center in orbit, but reliable low-power inference, storage buffering, radiation tolerance and power-aware workload scheduling.

Stage 2: Orbital cloud / distributed compute

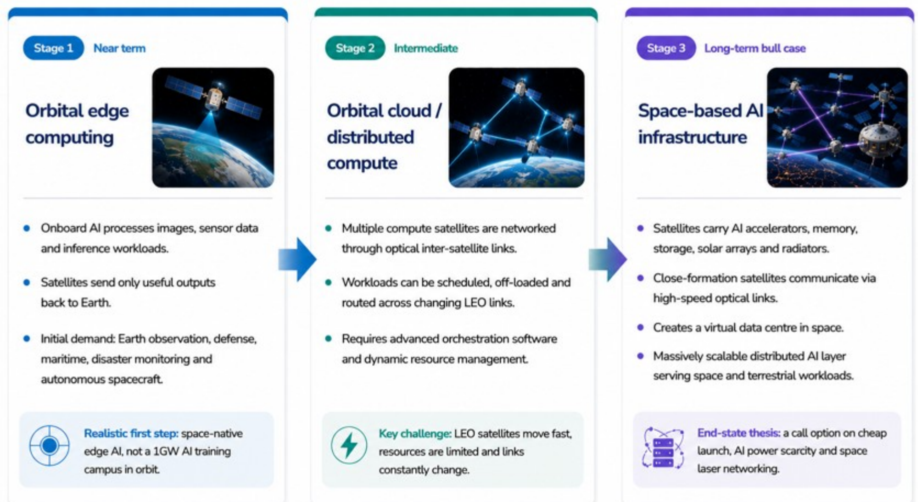
The intermediate stage links multiple compute satellites into a distributed orbital network. Workloads can be scheduled, off-loaded and routed across satellites as they move through LEO. The system starts to resemble a moving cloud region rather than a collection of independent spacecraft. The key capability is orchestration software that can manage changing satellite links, limited power, thermal headroom, distributed storage and compute availability in real time.

Stage 3: Space-based AI infrastructure

This is the ultimate objective. Orbital compute evolves beyond edge processing into an

AI infrastructure layer, with modular compute satellites carrying AI accelerators, solar arrays and radiators, connected through high-speed optical links. At this stage, the bottleneck shifts from terrestrial grid and cooling constraints to launch cost, kg/kW, radiation tolerance, radiator efficiency and autonomous maintenance. Google's Project Suncatcher represents one of the earliest public blueprints for this long-term vision.

Exhibit 4: Orbital Compute Roadmap



Source: Google Research; Orbital Edge Computing Survey; Morgan Stanley Research

Who are the key players?

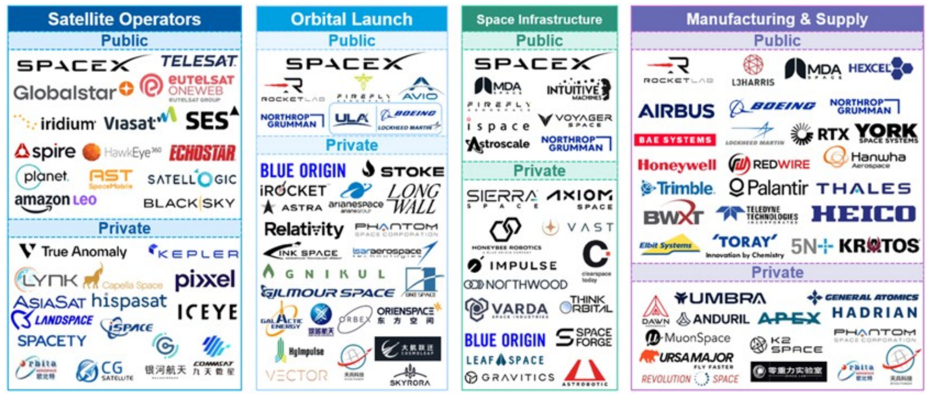
While NASA is an independent agency of the US federal government responsible for the United States' civil space program and for research in aeronautics and space, several companies are now driving space computing:

- **SpaceX (Starlink)** has accomplished numerous missions, including sending the first privately-funded spacecraft (Dragon) to dock with the International Space Station (ISS). They aim to reduce space transportation costs to enable colonization of Mars, which, if successful, could redefine human access to distant celestial bodies.

- **Eutelsat OneWeb** is a major LEO player backed by the UK government and strategic investors, focusing heavily on enterprise, maritime, and government communications.
- **Amazon's Project Kuiper** is the company's emerging LEO network, designed to integrate with its AWS cloud infrastructure. **Blue Origin** focuses on building reusable rockets that could ultimately enable millions of people to live and work in space.
- **China (Guowang)** is working on three major constellations: 1) Thousand Sails, based in Shanghai, which has already flown 750 satellites and has set a goal of reaching global coverage by 2027 and 15,000 by 2030; 2) a military network; and 3) a commercial network in the works – the latter of which are in the thousands of satellites. China is working on a different approach, focused on edge computing in space, where data (such as satellite imagery) is processed in orbit, instead of sending data back to Earth. It is funded, mandated and written into national policy

(unlike the US' private-led approach).

Exhibit 5: Global Public and Private Space Enablers



Note: Meant to give a representative snapshot. Does not capture 100% of exposed players. Does not include government agencies or government-owned enterprises. Infrastructure includes space stations, rovers, landers, ground stations, etc. Source: Company Websites, Morgan Stanley Research

The space computing economics

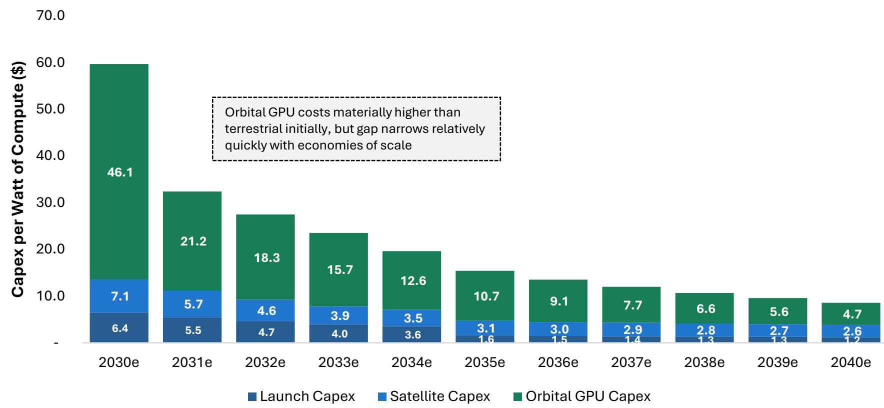
This section uses the MS SpaceX initiation report as the primary benchmark for orbital-compute economics. For details, please see SpaceX initiation – "AI's Final Frontier".

The space computing economics: cost/kg becomes cost/W. Orbital compute should not be framed as "free power and free cooling." Space offers abundant solar energy and avoids terrestrial land, water and grid constraints, but these are replaced by launch cost, satellite hardware, solar arrays, radiators, radiation tolerance and autonomous operations. We believe the right question is whether reusable launch and repeatable satellite manufacturing can drive orbital compute toward competitive cost per watt and faster time-to-power over the 2030s.The most detailed MS benchmark model divides orbital compute capex into three buckets: **launch capex, satellite hardware capex and compute payload capex.** In that framework, orbital compute capex/W declines as launch cost falls, satellite hardware scales and orbital GPU premiums narrow. The model assumes orbital compute capex/W falls from roughly **US$166/W in 2028** and **US$60/W in 2030** to **US $32/W** in 2031, **US$15/W** in 2035 and **US$9/W** by 2040. Assuming a five-year useful life, the 2031 estimate implies roughly **US$6.5/W/yea**r, broadly comparable with the current industry Blackwell benchmark of **US$6.8/W/**year.

Launch cost is necessary but not sufficient. A lower $/kg is the key enabler, but orbital compute also depends on launch cadence, compute mass per satellite, radiator efficiency, solar-array mass, satellite hardware cost, radiation loss and useful life. Our Embodied AI/ Robotics team models Starship launch cost falling toward **~US$500/kg** by 2030, below **US$200/kg** by 2035 and below **US$150/kg** by 2040, but also highlights that early Starship launches may deploy only single-digit MWs of compute. That makes launch cadence and time-to-power as important as pure cost parity.

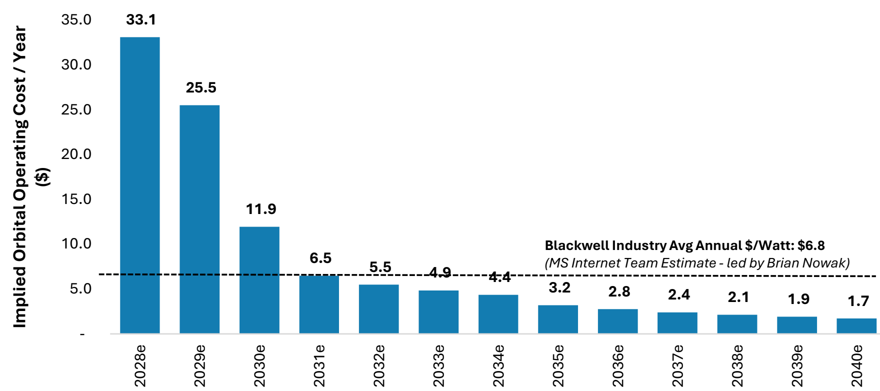
The conclusion is bullish but nuanced: orbital compute is unlikely to be cost-competitive at first, but could become strategically valuable if it becomes the fastest scalable source of AI compute capacity. In a constrained AI infrastructure market, customers may pay a premium for near-term access to large-scale compute even before orbital systems reach full cost parity. The long-term investment debate is therefore not only "*Is space cheaper?*" but also "*Can orbital compute scale faster than terrestrial capacity constrained by power, land, cooling and permitting?*"

Exhibit 6: Capex per Watt for Orbital Compute. Since the vast majority of orbital compute costs is depreciation, this figure can be divided by 5 years to derive a rough estimate of annual cost/Watt for compute deployed in any given year.



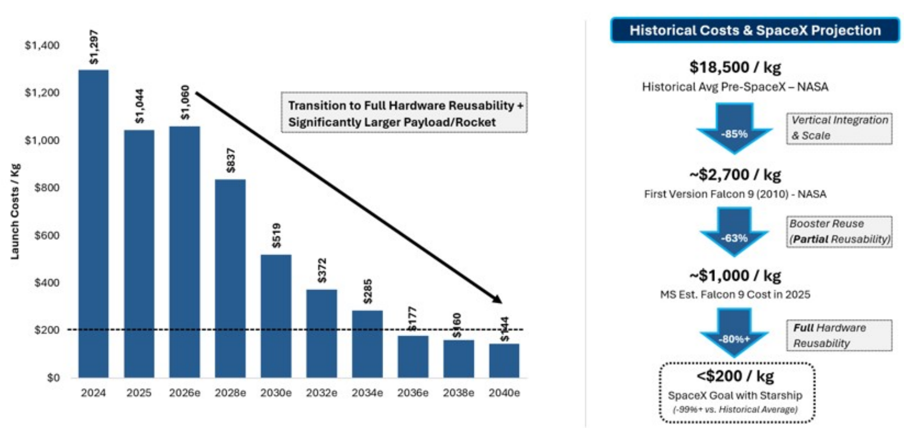
Note: Chart starts in 2030 as 2028-2029 figures are artificially high due to low volumes and economies of scale. Source: Morgan Stanley Research estimates

Exhibit 7: Our estimates imply orbital has a cost advantage vs. current industry terrestrial costs in 2031 (bars are orbital capex for newly deployed capacity divided by 5-year useful life, i.e., ongoing depreciation expense for new capacity, since the vast majority of ongoing orbital compute expenses is depreciation on upfront capex).



Note: For orbital compute, this uses newly deployed capacity in any given year, not avg compute in use. Source: Morgan Stanley Research estimates

Exhibit 8: Projected SpaceX Launch Costs/Kg Over Time (MSe)



Source: Company Data, NASA, Morgan Stanley Research estimates (e)

The compute payload

The more credible architecture is a purpose-built compute satellite, not a standard data-center rack launched into orbit. Our model divides an orbital compute satellite into three major systems: **thermal radiator, solar array, and bus / compute / other subsystems.** Early-generation AI satellites are modeled at roughly 150kW total power, 120kW compute draw and ~2.1 tons of satellite mass. Later generations scale toward 913kW total power, 730kW compute draw and ~6.1 tons, helped by larger launch vehicles, lighter solar arrays and more efficient radiators.

The key metric is compute per ton. In our model, compute density improves from roughly **56kW/ton** in early AI satellites to **80kW/ton** and eventually **120kW/ton**. This is why kg/kW, radiator area, solar-array efficiency and satellite hardware cost matter as much as launch cost. The orbital platform is not just carrying chips; it must also carry the power and thermal system that allows those chips to run.

The trade-offs

Falling launch costs: Reusable heavy-lift rockets are the key unlock. Google Research suggests that if launch prices fall below roughly US$200/kg, the cost of launching and operating space-based compute infrastructure could become more comparable with reported terrestrial data-center energy costs on a per-kilowatt/year basis. This is directionally consistent with our SpaceX initiation framework, but should not be read as full all-in data-center cost parity: orbital compute still requires satellite hardware, compute payloads, solar arrays, thermal radiators, radiation tolerance, autonomous operations and periodic replacement.

Time-to-power: Terrestrial AI data centers face grid interconnect, permitting, equipment and construction delays. Orbital compute could become valuable if it offers faster deployment of incremental compute, even before it reaches full cost parity. The key bottleneck is not just lower $/kg, but the ability to deploy enough compute per launch and reach high launch cadence.

Edge compute and AI bottlenecks: Satellites currently generate more data than transmission links can send back to Earth. In-orbit processing allows for real-time analysis of Earth observation, climate monitoring, and secure defense, utilizing inter-satellite free-space optical (laser) communication.

Bypassing grid congestion: Space computing allows tech companies to circumvent heavily congested terrestrial power grids and data sovereignty restrictions in specific geographic regions. Processing data in orbit may also support certain cross-border or space-native applications, although compliance with national data-sovereignty and regulatory requirements remains essential.

Solar exposure: No night cycles, atmospheric filtering, or weather interference. Orbital solar arrays can achieve capacity factors above 95%, reducing equivalent energy costs to fractions of a cent per kilowatt-hour, compared to 20-25% capacity factors for terrestrial solar.

Cooling: 40% of the power required to run a data center is used for cooling. But in space, the temperature is -270°C, which should make cooling much less of a problem.

Maintenance: Failed hardware cannot be repaired easily in orbit. Instead, orbital compute needs redundancy, autonomous fault detection, workload rerouting and planned replacement cadence. Our SpaceX model assumes a five-year useful life across orbital compute generations and includes a 4% annual compute radiation loss rate, or roughly 20% by end of life, which reinforces the need for spare capacity and regular satellite replacement.

Key challenges

The space industry is growing rapidly, but it is facing several challenges, including:

Cost of launching AI computing rack payloads into orbit: This has limited the growth of the space industry and its ability to bring down the cost of accessing space. As a result, many new companies are focusing on developing reusable spacecraft or alternative launch technologies to reduce the costs of transportation in space, such as electromagnetic propulsion or other methods of non-rocket space launch techniques.

Supply chain risks: The industry is vulnerable to supply chain bottlenecks, particularly due to a reliance on specialized materials and a limited number of global launch providers.

Regulatory: Regulatory complexities and international legislation surrounding space exploration can also create obstacles. Lack of traffic management and the absence of enforceable, globally agreed-upon space traffic management rules and "congestion pricing" creates major market failures.

Space debris and risk of collision: A growing amount of debris orbiting Earth could potentially damage satellites or even human habitats in space. The number of active satellites in orbit has increased rapidly over the past decade, from fewer than 2,000 in 2019 to more than 14,000 today, significantly increasing congestion in low Earth orbit. The International Space Station (ISS) gets radar checks three times a day to avoid debris comprising nearly 50,000 objects today (1,500 defunct satellites, 2000 second stage rocket bodies that weigh 1-8 tons, 10cm+ debris). Addressing the problem of orbital debris will be critical for ensuring sustainable growth within the space industry.

Thinking big

AI manufacturing on the Moon. This would involve local production of photovoltaic and radiators on the Moon, which would allow the mass to be made there, so avoiding the need to transport to the Moon from Earth. Because the Moon has no atmosphere and only one-sixth the Earth's gravity, AI satellites can be accelerated into deep space without a rocket. Payloads can be sent to space using electro-magnetic propulsion or linear electric motors and no longer expensive rockets.

Quantum networks. The development of quantum communication networks and a master network to unite them all will be one of the most important technological innovations of the 21st century. Satellite-based quantum communication is known as the most feasible way to achieve global quantum communication. There is a world that is invisible to us where nature behaves very differently at the atomic scale, and particles can exist in multiple states at the same time, can be entangled or share information with one another, over very long distances even without a physical connection. By transferring information via satellites using individual particles of light (photons), a quantum internet would be capable of sending enormous amounts of data over vast distances, instantaneously and privately. See *Asia Technology Disruption Decoded: The Rise of Quantum Networks (16 Dec 2020).*

How it works – the orbital compute stack

Orbital compute is not a terrestrial data center rack bolted onto a satellite. It is an integrated space system that combines an AI server, satellite bus, solar power system, thermal-management system, optical network and autonomous operations software. The core design challenge is to make compute useful in orbit while managing five physical constraints: **mass, power, heat, radiation and serviceability.**

The more credible architecture is modular rather than monolithic. Investors should not picture a one-million-square-foot data center assembled in Low Earth Orbit (LEO). Instead, the likely path is many smaller compute satellites or orbital compute modules, each with its own power, thermal and networking system, linked together through optical inter-satellite laser links. In this model, the "data center" is virtual: compute is distributed across satellites, while software manages where data is processed, stored and routed.

Two design choices sit at the center of this architecture: where the compute satellites fly and how they communicate with each other. **The orbit determines solar exposure, battery needs, radiator geometry and latency. The optical network determines whether satellites remain isolated edge nodes or become a distributed compute fabric.**

Google's Project Suncatcher is the clearest public reference architecture: solar-powered satellites carrying TPUs, operating in dawn-dusk Sun-Synchronous LEO and connected through free-space optical inter-satellite links. Google also argues for modular arrays of smaller satellites rather than monolithic space data centers because giant structures introduce assembly, collision-avoidance and mass challenges.

Exhibit 9: Orbital compute system stack

Layer	What it does	Key technology requirements
AI compute payload	Runs inference, image processing, sensor analytics and selected AI workloads in orbit	AI accelerators, CPUs, High Bandwidth Memory, error correction, fault-tolerant software
Server / payload module	Packages chips, boards, storage and power delivery into a launch-tolerant compute unit	Rugged AI trays, vibration tolerance, remote management, modular replacement
Power system	Converts solar energy into stable compute power	Solar arrays, batteries, high-efficiency direct-current conversion, power management integrated circuits
Thermal system	Moves heat from chips to radiators and rejects it into space	Cold plates, heat pipes, thermal interface materials, radiator coatings, thermal sensors
Orbit and formation	Determines solar exposure, latency, radiator geometry and satellite spacing	Dawn-dusk Sun-Synchronous Orbit, close-proximity formation, collision avoidance, autonomous control
Optical network	Connects satellites to each other and to ground/cloud systems	Laser terminals, coherent optics, beam steering, optical digital signal processors, Dense Wavelength Division Multiplexing
Storage and orchestration software	Buffers data, schedules workloads, routes traffic and handles failures	Solid-state drives, controllers, task scheduling, telemetry, cybersecurity, autonomous operations

Source: Morgan Stanley Research

1. Compute payload: from satellite sensor to AI node

The first redesign is the satellite payload itself. Traditional satellites are primarily sensors and communications nodes: they collect imagery or signals, downlink raw data and leave most processing to ground-based systems. Orbital compute turns the satellite into an edge AI node. The satellite can process imagery, sensor feeds or inference workloads onboard, then send only the useful output back to Earth.

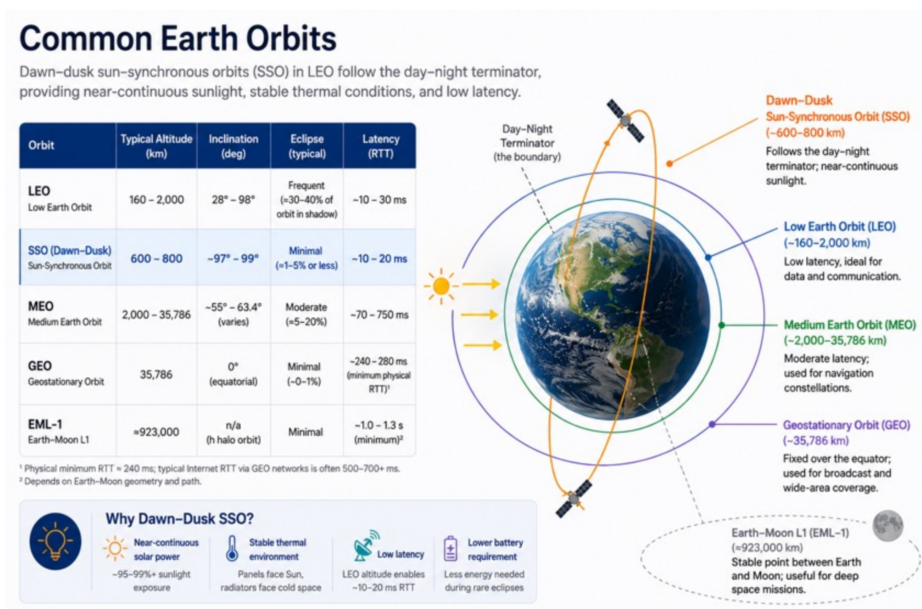
This matters because space-to-ground bandwidth is limited and expensive. For Earth observation, a satellite could identify clouds, ships, fires, military assets or disaster zones before down linking results. For defense and disaster monitoring, the value is faster time-to-insight. The compute requirement at this stage is not a full AI training cluster in orbit; it is reliable low-power inference, storage buffering, radiation tolerance and power-aware workload scheduling.

2. Orbit and power: sun synchronous orbit is part of the architecture

The preferred orbit for many orbital-compute concepts is dawn-dusk Sun-Synchronous Orbit (SSO), a LEO that follows the day-night terminator. The satellite remains close to continuous sunlight, which is attractive for solar-powered compute. It is valuable because the solar wings can generate power for longer periods and batteries can be smaller. The radiator can be positioned away from the Sun to reject heat into cold space. This is why dawn-dusk SSO is attractive for orbital compute: it combines LEO latency with more stable solar power and thermal conditions than many other orbital profiles.

This does not make power free. The system still needs solar-array area, battery mass, high-efficiency power electronics, power conversion and fault protection. The power budget directly affects how much compute can run, how long it can run and how much redundancy the system needs.

Exhibit 10: Common orbit choices – why dawn-dusk SSO is the sweet spot for compute



Source: ESA/NASA orbit references; Morgan Stanley Research.

3. Thermal system: cooling becomes a radiator problem

Space avoids terrestrial water cooling, cooling towers and air-based cooling systems, but it does not eliminate thermal management. In vacuum, there is no air convection, so heat from AI chips must be moved through cold plates, heat pipes or pumped loops to radiators, then rejected into space through radiation.

This is one of the most underappreciated bottlenecks. AI accelerators are dense heat sources, and the system must move heat away from the chips without a technician, chiller plant or normal data-center airflow. Google identifies thermal management as a critical challenge for power-dense TPUs operating in vacuum, requiring thermal interface materials and heat transport to move heat from chips to radiator surfaces.

The key metric changes from Power Usage Effectiveness (PUE) in terrestrial data centers to space-specific metrics such as kilograms per kilowatt, radiator area per kilowatt and heat-rejection efficiency.

4. Optical networking: laser links become the data-center fabric

Inter-satellite laser links are the space equivalent of data-center fiber. They allow satellites to move data between nodes without routing everything through ground stations. This is what turns orbital compute from isolated edge processors into a distributed compute fabric.

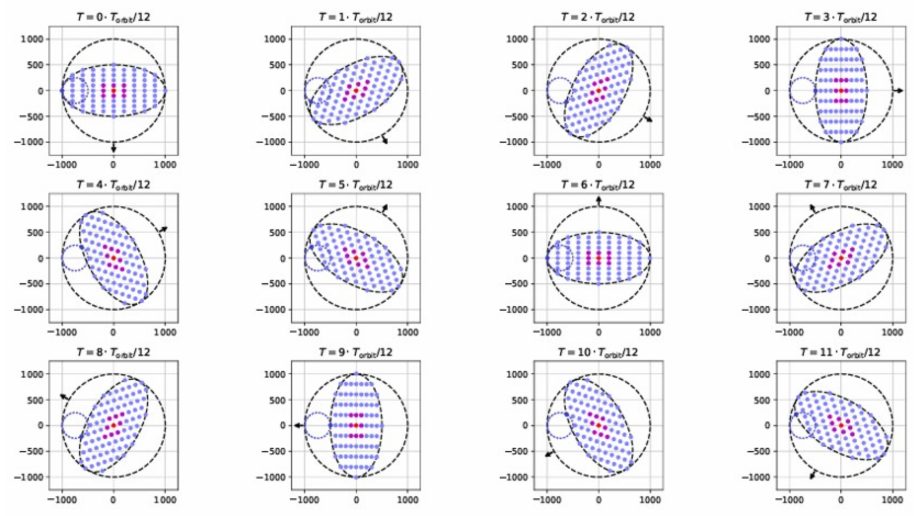
For Stage 1, networking requirements are manageable because each satellite can process data locally and downlink useful outputs. For Stage 2 and Stage 3, the requirements become much harder. Multiple satellites must share workloads, route data and maintain connectivity while moving quickly through LEO. That requires high-bandwidth optical links, precise pointing, dynamic routing and orchestration software.

Existing satellite laser links (such as Starlink) validate the direction of travel, but AI-scale distributed compute would require a step-change in bandwidth and coordination. Google notes that commercial optical inter-satellite links are typically in the 1-100Gb per second range, while tightly coupled machine-learning clusters may require aggregate bandwidth per link on the order of 10Tb per second. Optical networking is therefore not a secondary feature; it is the core unlock that determines whether orbital compute becomes a true distributed cloud or remains a collection of isolated satellites.

5. Formation flight: distance becomes a bandwidth variable

In terrestrial data centers, servers are fixed inside a building and connected by fiber. In orbital compute, the compute nodes are satellites moving at orbital velocity. Their relative distance and formation matter because optical-link performance improves when satellites are closer together. This is why Stage 2 and Stage 3 require more than normal satellite constellation management. The system may need close-proximity satellite formation, precision pointing, autonomous control and collision avoidance.

Exhibit 11: Google's illustrative system: 81-satellite cluster in a 1km radius at a mean altitude of 650 km, with nearest-neighbor distances varying roughly between 100-200m



Source: Google Research.

6. Storage and orchestration: software becomes the control plane

The final layer is software. In terrestrial cloud, workloads are scheduled across relatively stable data centers and networks. In orbital compute, the "cloud region" is moving. Satellites change position, power availability, thermal headroom, link quality and ground-station access over time.That creates a new orchestration challenge. The system needs to decide which satellite should process which workload, whether data should be processed locally or off-loaded to another satellite, when to downlink results, how to handle errors and how to operate through partial failures. Storage is also important because satellites may need to buffer data until they have enough compute capacity, link availability or ground-station visibility.

What has to be redesigned?

Orbital compute is not simply a terrestrial data center moved into orbit. It requires a redesign of the full engineering stack. Grid power and power purchase agreements are replaced by solar arrays, batteries and power management; chillers and liquid cooling are replaced by heat pipes and radiators; fiber networks are replaced by optical inter-satellite laser links; and human technicians are replaced by redundancy, autonomous operations and modular replacement. The key efficiency metrics also change: instead of Power Usage Effectiveness, orbital compute is likely to be measured by cost/kg, compute W/kg, capex/W, radiator efficiency and useful life.

Exhibit 12: What has to be redesigned?

Terrestrial AI Data Center Stack		Orbital Compute Equivalent
Grid power / Power Purchase Agreements	→	Solar arrays + batteries + power management
Chillers / liquid cooling / water	→	Heat pipes + radiators
Fiber and switches	→	Optical inter-satellite laser links
Technicians	→	Redundancy, autonomous operations, modular replacement
Data Center Building	→	Satellite bus / orbital platform
Cloud region	→	Moving Low Earth Orbit compute constellation
Power Usage Effectiveness	→	Kilograms per kilowatt (kg/kW), launch cost per kilogram ($/kg), radiator efficiency

Source: Morgan Stanley Research

Mapping the technology supply chain

Orbital compute systems rely on a complex web of ground infrastructure, including gateway stations, antenna arrays, network operations centers, terrestrial fiber backhaul connections, and data routing systems. This infrastructure represents the physical interface between orbital networks and the terrestrial internet, and without it, the satellites above are essentially useless. Space data centers require radiation-hardened components, physical redundancy, and exotic shielding, which significantly drives up base costs.

Direct operators vs. listed supply-chain enablers. SpaceX is the most important direct reference platform for orbital compute, but it should be separated from the listed tech supply-chain map. The reason is simple: SpaceX is a vertically integrated operator spanning launch, satellite manufacturing, Starlink connectivity and AI infrastructure, while our stock table is designed to identify public-market suppliers enabling the broader orbital-compute stack.

Many pure-play operators remain private or sit inside mega-cap platforms, while near-term revenue is unlikely to be material for most public companies. The map therefore separates the **AI infrastructure payload** from the **orbital overlay**. The core AI payload is familiar: accelerators, adaptive SoCs, custom silicon, HBM, DRAM, foundry, high-reliability PCBs and advanced components. The orbital overlay adds optical links, RF front-ends, phased-array terminals, radiation-tolerant semiconductors, high-reliability passives, power conversion, batteries, solar cells, thermal management and satellite communications hardware.

We group the stock universe into six layers:

1. **AI silicon, memory and custom compute.** This is the core payload layer. Orbital compute still needs accelerators, adaptive SoCs, custom silicon, foundry capacity, HBM and resilient memory. Relevant names include NVIDIA, AMD, Broadcom, Micron, TSMC, SK hynix and Samsung.
2. **High-reliability components, PCBs, passives and system integration.** Space systems need components that can survive vibration, thermal cycling, radiation exposure and long periods without human service. This creates read-through to MLCCs, passives, high-reliability PCBs and system integration. Relevant names include Murata, Taiyo Yuden, TDK, Samsung Electro-Mechanics, TTM Technologies, Compeq, Unitech and Hon Hai. This bucket is important because orbital compute modules must be packaged as launch-tolerant, remotely managed systems rather than standard terrestrial racks.
3. **Optical links, photonics & terminals.** Optical inter-satellite links are the space equivalent of data-center fiber. They determine whether orbital compute remains a set of isolated edge nodes or evolves into a distributed compute fabric. Relevant names include Coherent, Lumentum, Eoptolink and CACI. AI-scale orbital compute would require higher bandwidth, tighter pointing, better beam steering and more dynamic routing than conventional satellite communications.
4. **RF, phased-array terminals and satellite communications hardware.** This bucket captures the connectivity layer between satellites, ground stations and user terminals. Relevant technologies include RF front-ends, phased-array antennas,

satellite terminals, NTN chipsets, microwave modules, connectors and ground-station hardware. Relevant names include STM, Qorvo, Wistron NeWeb, Universal Microwave Technology, Intellian Technologies, Gilat Satellite Networks, Filtronic, Amphenol, WinSemi, MediaTek and Rapidtek Technologies.

5. **Radiation-tolerant and space-grade semiconductors.** Space is a harsher environment than terrestrial data centers. Radiation can cause bit flips, calculation errors, device degradation and system faults. Orbital compute therefore needs radiation-tolerant signal chain, memory, FPGAs, control electronics, RF devices and power management. Relevant names include Analog Devices, Infineon, Microchip Technology, Mercury Systems and Shanghai Fudan. This bucket is critical because the hardware must operate for years with limited or no physical maintenance.
6. **Space power, batteries and thermal management.** Power and heat define the practical ceiling for compute in orbit. Orbital compute replaces terrestrial grid power and water cooling with solar arrays, batteries, DC power conversion, thermal transport, radiators and power-aware workload management. Relevant names include Redwire, GS Yuasa, Sharp, Lite-On, Umicore, LG Energy Solution, AXT and Delta Electronics. This layer is important because the economics of orbital compute depend less on conventional data-center Power Usage Effectiveness and more on kilograms per kilowatt, launch cost per kilogram, battery mass and radiator efficiency.

Exhibit 13: Orbital compute technology supply chain

Company	Ticker	Region	Rating	Last Close (LC)	Market Cap ($mn)	Key orbital read-through
1. AI silicon, memory & custom compute						
NVIDIA	NVDA	US	OW	195	4,714,886	Space-1 / Vera Rubin, Jetson Orin and IGX Thor make NVIDIA the default orbital AI accelerator reference.
AMD	AMD	US	EW	518	844,357	Space-grade Versal adaptive SoCs and data acceleration for onboard processing payloads.
Broadcom	AVGO	US	OW	360	1,714,870	Custom silicon and networking read-through for AI payloads, ground gateways and high-bandwidth systems.
Micron	MU	US	OW	976	1,101,788	HBM / DRAM supply exposure for AI compute payloads and resilient orbital memory.
TSMC	2330 TT	Taiwan	OW	2,445	1,982,637	Leading-edge foundry for orbital AI silicon; any scaled compute layer ultimately depends on advanced logic supply.
SK hynix	000660 KS	Korea	OW	2,425,000	1,123,802	HBM is a binding constraint for any gigawatt-scale deployed compute architecture.
Samsung Electronics	005930 KS	Korea	OW	309,500	1,317,860	HBM / DRAM exposure; additional satellite communications optionality through modem / RF ecosystem.
2. High-reliability components, PCBs, passives & system integration						
Murata	6981 JP	Japan	OW	11,080	125,051	Space-grade MLCC leader; high-reliability passives for satellites, power modules and payload electronics.
Taiyo Yuden	6976 JP	Japan	UW	20,560	15,953	High-reliability MLCCs and passive components for harsh-environment electronics.
TDK	6762 JP	Japan	OW	3,699	43,533	Passives and electronic components for space, power and satellite systems.
Meiko Electronics	6787 JP	Japan	EW	29,330	4,668	Strong market position in PCB antennas for LEO satellite communication receivable antennas, and could potentially expand into satellite-grade PCB
Samsung Electro-Mechanics	009150 KS	Korea	OW	1,989,000	98,314	MLCC / substrates exposure for satellite electronics and power modules.
TTM Technologies	TTMI	US	NC	156	16,199	PCBs for high-reliability aerospace, defense and compute payload electronics.
Hon Hai	2317 TT	Taiwan	OW	241	105,488	EMS / system-integration optionality plus direct LEO satellite communications hardware validation through Foxconn PEARL satellites.
Compeq	2313 TT	Taiwan	NC	230	8,794	Advanced PCB exposure to SpaceX satellite, ground receiving stations and terminals.
Unitech	2367 TT	Taiwan	NC	55	1,215	Advanced PCB exposure to SpaceX user terminals/ground stations.
3. Optical links, photonics & terminals						
Coherent	COHR	US	EW	333	65,218	Laser optics and photonics read-through for optical inter-satellite links and ground optical systems.
Lumentum	LITE	US	EW	728	56,663	Laser optics and photonics exposure to high-bandwidth optical networking.
CACI	CACI	US	NC	503	11,108	Optical communications terminals and space communications systems; closer to the orbital laser-link bottleneck than generic IT services.
Eoptolink	300502 CH	China	OW	526	108,158	Optical transceiver read-through for high-bandwidth orbital networking.
4. RF, phased-array terminals & satellite communications hardware						
STMicroelectronics	STM	Europe	NC	68	60,749	RF front-end / BiCMOS and space-grade electronics exposure for Starlink-style phased-array antennas, LEO terminals and satellite communications.
Filtronic	FTC LN	Europe	NC	2	699	High-frequency Solid State Power Amplifiers (SSPAs) installed in ground stati...
Gilat Satellite Networks	GILT	Israel / US	NC	13	974	Ground systems and satellite communications terminal exposure.
Qorvo	QRVO	US	EW	88	7,725	RF products and semis for satellite communications and phased-array systems.
Amphenol	APH	US	NC	165	202,484	High-reliability connectors, interconnects and cable assemblies for satellites, ground infrastructure and launch systems.
MediaTek	2454 TT	Taiwan	OW	4,195	209,370	5G NTN / satellite IoT chipset exposure; relevant to direct-to-device and satellite communications hardware rather than AI compute payloads.
WinSemi	3105 TT	Taiwan	UW	411	5,448	Taiwan compound-semiconductor foundry with GaAs / GaN RF exposure; potential beneficiary of LEO satellite communications, higher-frequency
Wistron NeWeb	6285 TT	Taiwan	NC	258	3,902	LEO ground terminals, phased-array antennas and satellite communications equipment.
Universal Microwave Technology	3491 TT	Taiwan	NC	1,340	2,883	RF / microwave modules for satellite terminals and phased-array communications.
Rapidtek Technologies	6980 TT	Taiwan	NC	136	123	RF, satellite applications and ground-to-space / LEO terminal exposure; potential Taiwan orbital-communications overlay name.
Intellian Technologies	189300 KS	Korea	NC	73,700	493	Ground-station antennas and satellite terminal exposure.
5. Radiation-tolerant / space-grade semiconductors						
Infineon	IFX-DE	Europe	NC	77	115,294	Radiation-hardened power management, memory/RF and high-reliability power conversion for satellite platforms and compute payload power rails.
Analog Devices	ADI	US	OW	377	183,710	Radiation-hardened signal-chain, power and control electronics for space-qualified systems.
Microchip Technology	MCHP	US	EW	85	45,882	Rad-hardened semiconductors, FPGAs and space-qualified control electronics.
Mercury Systems	MRCY	US	NC	126	7,578	RF products, semiconductors and mission electronics for space and defense platforms.
Shanghai Fudan	1385 HK	China	OW	31	3,247	China space-grade FPGA supplier, benefiting from China's low earth orbit satellite launches
6. Space power, batteries & thermal management						
Redwire	RDW	US	NC	11	2,701	Deployable solar arrays (ROSA), structures and space infrastructure hardware; higher-beta pure-play exposure.
AXT	AXTI	US	NC	57	3,704	Germanium substrate exposure with China supply-chain risk framing.
GS Yuasa	6674 JP	Japan	OW	6,737	4,190	Space lithium-ion cells and commercial satellite battery exposure.
Sharp	6753 JP	Japan	NC	639	2,573	Space-grade triple-junction solar cell exposure.
Delta Electronics	2308 TT	Taiwan	OW	2,075	168,540	Power conversion, DC/DC modules and power-management exposure for orbital compute power rails and ground infrastructure.
Lite-On	2301 TT	Taiwan	EW	219	15,528	Power supply and power-conversion exposure for LEO satellite / ground infrastructure power systems.
LG Energy Solution	373220 KS	Korea	EW	362,500	55,499	Battery exposure for satellite and space-power applications.

Source: Factset, Company Data, Morgan Stanley Research; NC = companies not covered by Morgan Stanley Research

Valuation Methodology and Risks

Infineon Technologies AG (IFXGn.DE)

We apply a 50x P/E multiple to the server rack contribution as we think the structural driver deserves a premium to the rest of the group. We continue to view the rest of the group as a cyclical story valuing it at 24x reflecting our view that Infineon is moving through a cycle recovery. Using a SOTP and assuming a 10% WACC, we value IFX at €91.

Risks to Upside

- Faster than expected recovery of industrial output & consumer sales
- Longer autos semi cycle than normal & better pricing power with OEMs
- Greater deployment of power semis to the data centre than expected

Risks to Downside

- Sustained weakness in global automotive sales and/or PMIs
- Capex ramp leads to over-capacity and impairs future returns
- Backlog slowdown
- Chinese designs limiting longer term share expansion

STMicroelectronics NV (STMPA.PA)

We value STM at €78/share based on a sum-of-the-parts framework. We value the underlying core business at 18x P/E multiple on FY28e EPS €2.11 and the optical and LEO business on a 36x P/E multiple on FY28e EPS €1.34, to reflect the different growth trajectories of the cyclical and structural segments. Discounting using a 10% WACC, this implies a €78 PT.

Risks to Upside

- Market adoption of SiC
- Design wins growth across autos and industrials
- Larger than expected optical ramp in 2H26/FY27
- LEO sales materially beyond $3bn cumulative FY26/28

Risks to Downside

- Prolonged global macro weakness, drives weak autos & industrial OEM volumes
- MCU pricing weakness as inventory channel corrects
- Design socket loss/decreasing sensor content in key smartphones
- DRAM shortages impact auto sales

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(as of June 30, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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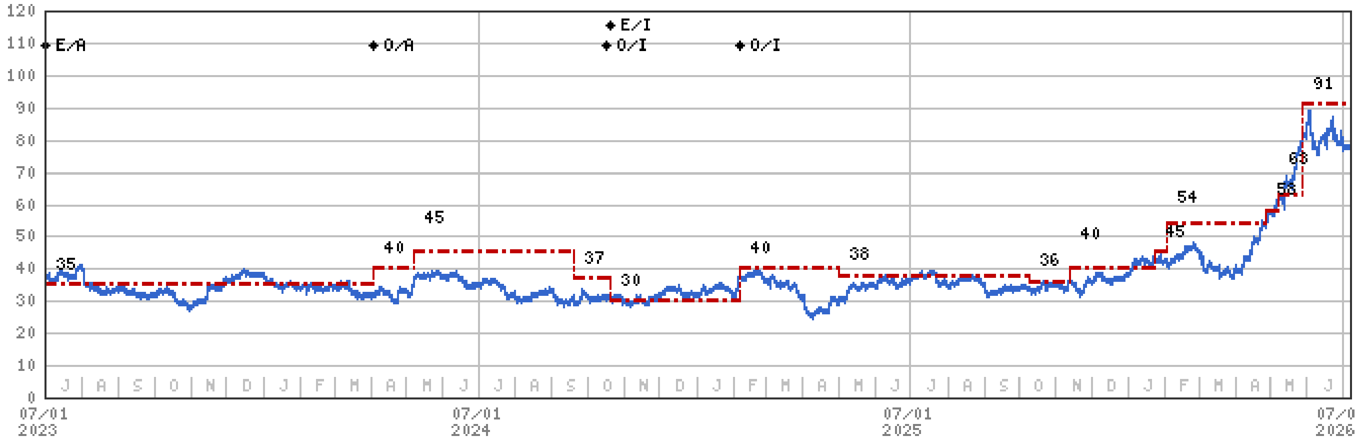
Stock Price, Price Target and Rating History (See Rating Definitions)

Infineon Technologies AG (IFXGn.DE) - As of 07/07/26 GMT in EUR
Industry : Technology - European Semiconductors



Stock Rating History: 7/1/21 : O/I; 12/8/21 : E/I; 2/8/22 : NA/I; 11/1/22 : NA/NR; 11/7/22 : E/A; 4/3/24 : O/A; 10/17/24 : O/I; 10/20/24 : E/I; 2/6/25 : O/I

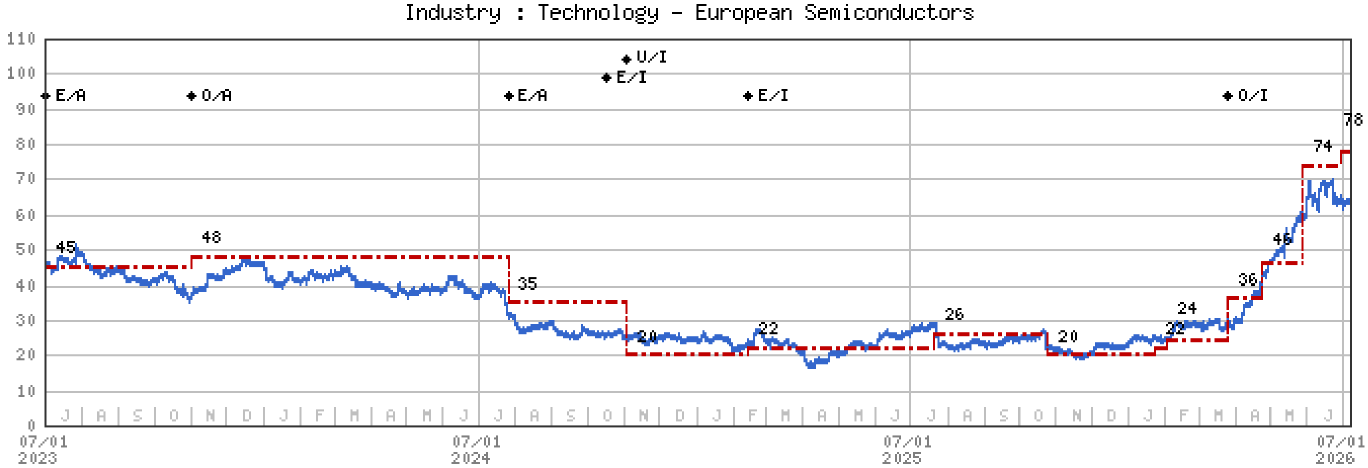
Price Target History: 4/15/21 : 38; 8/3/21 : 39; 11/10/21 : 42.5; 12/8/21 : 43; 2/8/22 : NA; 11/7/22 : 32.5; 1/31/23 : 35; 4/3/24 : 40; 5/8/24 : 45; 9/20/24 : 37; 10/20/24 : 30; 2/6/25 : 40; 5/2/25 : 38; 10/9/25 : 36; 11/12/25 : 40; 1/23/26 : 45; 2/2/26 : 54; 4/27/26 : 58; 5/7/26 : 63; 5/28/26 : 91

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

STMicroelectronics NV (STMPA.PA) - As of 07/07/26 GMT in EUR
Industry : Technology - European Semiconductors



Stock Rating History: 7/1/21 : O/I; 12/8/21 : E/I; 2/9/22 : NA/I; 11/1/22 : NA/NR; 11/7/22 : E/A; 11/1/23 : O/A; 7/26/24 : E/A; 10/17/24 : E/I; 11/3/24 : U/I; 2/13/25 : E/I; 3/26/26 : O/I

Price Target History: 4/29/21 : 39; 8/6/21 : 40; 10/28/21 : 43.5; 12/8/21 : 44.5; 1/27/22 : 46; 2/9/22 : NA; 11/7/22 : 36; 1/30/23 : 45; 11/1/23 : 48; 7/26/24 : 35; 11/3/24 : 20; 2/13/25 : 22; 7/21/25 : 26; 10/24/25 : 20; 1/23/26 : 22; 2/2/26 : 24; 3/26/26 : 36; 4/24/26 : 46; 5/28/26 : 74; 6/30/26 : 78

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
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